

STRENGTH OF MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for the right answer and 1 mark is allotted for the correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

- (i) The stress at which elongation of a material is quite large as compared to the increase in load is known as-
(a) Ultimate point (b) Yield point (c) Elastic limit (d) Rupture point.
- (ii) For a Poisson's ratio 0.4 for a material, the ratio of the shear modulus of elasticity to modulus of elasticity is- (a) 14/5 (b) 5/7 (c) 7/5 (d) 5/14.
- (iii) Which of the following increases bending stress in a beam?
(a) Increase in moment of inertia (b) Increase in section modulus (c) Decrease in distance from neutral axis (d) Decrease in moment of inertia.
- (iv) If a simple supported beam is loaded with a concentrated load W at the centre of the span then its B.M diagram takes the shape of -
(a) Rectangle (b) Parabola (c) Triangle (d) Square.
- (v) When a beam is subjected to a negative bending moment-
(a) Top fiber is in tension and bottom fiber is in compression (b) Top fiber is in compression and bottom fiber is in tension. (c) Minimum stress in top and bottom fiber (d) Both (a) and (b).
- (vi) The M.O.I. of a circle of diameter D with respect to the axis which is perpendicular to its diameter and passing through the centre is -
(a) $\frac{\pi}{64} D^4$ (b) $\frac{\pi}{32} D^4$ (c) $\frac{\pi}{32} D^3$ (d) $\frac{\pi}{16} D^3$
- (vii) In a simple supported beam carrying UDL ' w ' per unit length, the point of contraflexure -
(a) Lies in the centre of the beam (b) Lies at the ends of the beam (c) Depends upon the length of the beam (d) Does not exist.
- (viii) If a shaft diameter is doubled, its torsional strength increases by-
(a) 2 times (b) 4 times (c) 8 times (d) 16 times.
- (ix) The buckling load of a column for a given material depends upon-
(a) Poisson's ratio and slenderness ratio (b) Slenderness ratio and modulus of elasticity (c) Poisson's ratio and modulus of elasticity (d) Slenderness ratio and cross sectional area.
- (x) The ratio of torsional strengths of solid to hollow shafts, both having outside diameter D and hollow having inside diameter $D/2$, is-
(a) 15/16 (b) 1/8 (c) 1/16 (d) 16/15.

- (xi) In the context of beam bending, flexural rigidity is -
 (a) A measure of the beam's resistance to shear stress (b) A measure of the beam's resistance to bending
 (c) A measure of the beam's ability to support compressive loads (d) A measure of the beam's ability to support tensile loads.
- (xii) A column is fixed in one end and other end is free and has a radius of gyration $K = 2.5\text{ cm}$. The actual length of the column is 1.2 m . The slenderness ratio of the column is -
 (a) 96 (b) 80 (c) 90 (d) 120.
- (xiii) If the flexural rigidity of a beam becomes double then the deflection of the beam-
 (a) Increases 2 times (b) Becomes half (c) decreases 4 times (d) remains same.
- (xiv) Four springs each having stiffness k are arranged in series, then the equivalent stiffness will be-
 (a) $4k$ (b) $k/2$ (c) $2k$ (d) $k/4$.
- (xv) Which of the following end condition gives the lowest Euler's buckling strength for a column?
 (a) Both ends fixed (b) Both ends pinned (c) One end fixed, other free (d) One end fixed, other hinged.

GROUP-B

2. (a) Draw the stress-strain curve for M.S material and discuss salient points in the diagram. (3)
 (b) A circular steel bar having three segments is subjected to various forces at different cross sections as shown in fig. 1. The diameter of the first, middle and last section of the bar is 30 mm , 25 mm and 30 mm respectively. Determine the total elongation of the bar. Take $E = 200\text{ GPa}$. [3+5]

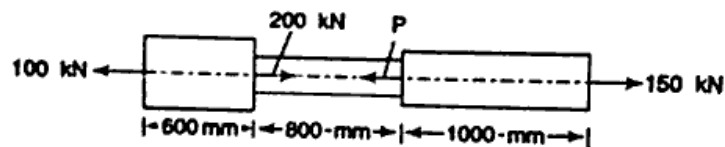


Fig. 1.

3. (a) A bar of 12 mm diameter and 2 m length is subjected to an axial load of 20 kN . The change in diameter is measured as 0.003 mm . If the modulus of rigidity of the bar is 80 GPa , determine (i) Poisson's ratio (ii) modulus of elasticity and bulk modulus.
 (b) No stress is developed in the rails when they are laid at 24°C . If the rails are 32 m long, calculate the stress developed in the rails at 80°C when it is allowed to expand 8 mm . Take co-efficient of linear expansion, $\alpha = 11 \times 10^{-6}/^\circ\text{C}$ and $E = 205\text{ GPa}$. [5+3]
4. Determine the moment of inertia of the composite shape given in the fig. 2 about x-axis and y-axis. [8]

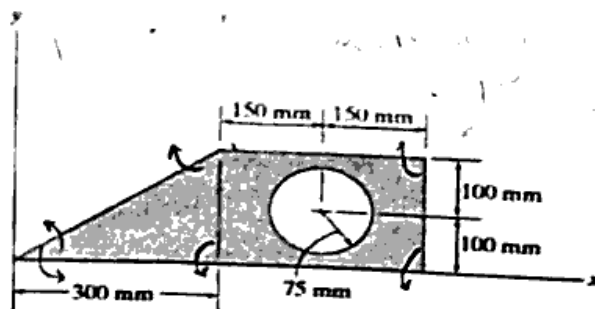


Fig. 2.

5. Draw shear force and bending moment diagrams for the beam as shown in the fig.3. Determine the maximum bending moment and its location. [8]

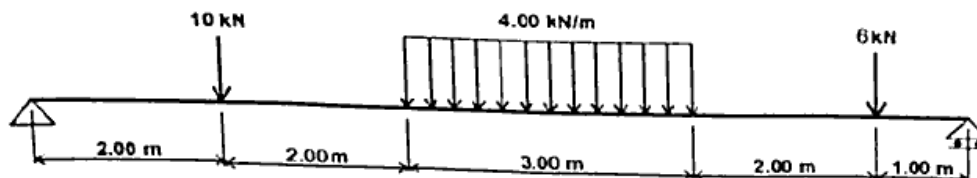


Fig.3.

- 6.(a) What do you mean by section modulus of a beam? (2)
 (b) A cast iron beam of T section as shown in fig.4 is simply supported on a span of 8m. The beam carries a uniformly distributed load of 1.5kN/m length on the entire span. Determine the maximum tensile and maximum compressive stress. [2+6]

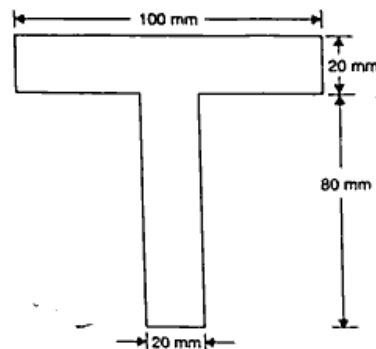


Fig.4.

7. (a) State the four assumptions in theory of bending.
 (b) A beam of uniform rectangular section 20cm wide and 30cm deep is simply supported at its ends. It carries a uniformly distributed load of 9 kN/m run over the entire span of 5 m. If the value of E for the beam material is $1 \times 10^4 \text{ N/mm}^2$, find the slope at the supports and maximum deflection. [2+6]
8. (a) What is torsional stiffness?
 (b) A hollow shaft with inner to outer diameter ratio $3/5$ is required to transmit 450kW at 120rpm with a uniform twisting moment. The shear stress in the shaft must not exceed 60 MPa and twist in a length of 2.5m must not exceed 1° . Calculate the least external and internal diameter of the shaft satisfying these conditions. Take $G = 80 \text{ GPa}$. [2+6]
9. (a) Draw the effective lengths of a column for various end conditions. (2)
 (b) A hollow cylindrical column of length 6m having external diameter 200mm and thickness 25mm is fixed at both ends. Determine the ratio of Euler's critical load to Rankine's critical load. Take $E = 80 \text{ GPa}$; allowable stress as 550 MN/m^2 and Rankine's constant as $1/1600$. [3+5]
10. (a) What are the different types of springs? State the functions of the springs.
 (b) A closely coiled helical spring is to carry a load of 500 N. The spring index is 10. If the maximum shear stress in the material of the spring is to be 80 MPa, determine mean coil diameter and wire diameter of the spring. If the stiffness of the spring is 20 N per mm deflection, calculate the energy stored in the spring and the number of turns in the spring. Take $G = 80 \text{ GPa}$ [2+6]